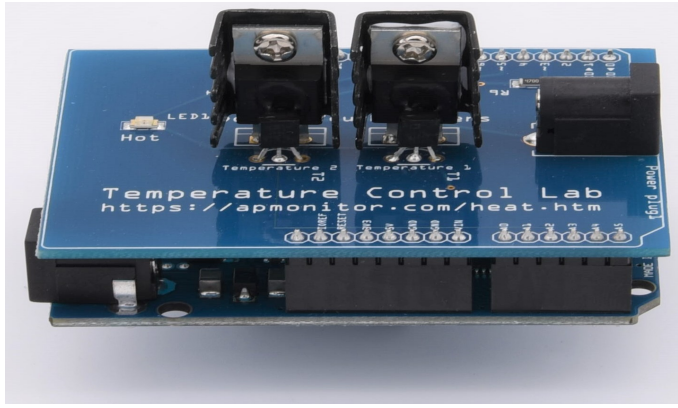

Figure 1: Temperature Control Lab (TCL) setup

1 Mechanistic Modeling of Temperature Control Lab (TCL)

In this section, the details of the 1st principle/mechanistic model of TCL have been studied, which forms the basis of this work. Later the model is used to perform simultaneous EKF and subsequently, this adaptive feature is integrated into the NMPC algorithm to perform simulation and experimental validation. .

The mechanistic model is developed using the energy balance equation for each of the two heaters. The convective and radiative heat exchange of individual heaters with surrounding and with themselves is taken into consideration. The model is represented by Equations 1 to 4.

$$Q_{C12} = UA_s(T_2 - T_1) \tag{1}$$

$$Q_{R12} = \in \sigma A(T_2^4 - T_1^4) \tag{2}$$

$$mc_p\frac{dT_1}{dt} = UA(T_\infty - T_1) + \in \sigma A(T_\infty^4 - T_1^4) + Q_{C12} + Q_{R12} + \alpha_1 Q_1 \tag{3}$$

$$mc_p\frac{dT_2}{dt} = UA(T_\infty - T_2) + \in \sigma A(T_\infty^4 - T_2^4) - Q_{C12} - Q_{R12} + \alpha_2 Q_2 \tag{4}$$

Where,
T_∞ = Ambient temperature
$Q_1 = 1^{st}$ Heater output
$\alpha_1 = 1^{st}$ Heater factor
$Q_2 = 2^{nd}$ Heater output
$\alpha_2 = 2^{nd}$ Heater factor
c_p = Specific heat capacity of the heater
A = Surface area not between the heaters
A_s = Surface area between the heaters

m = Mass of the heater
U = Overall heat transfer coefficient
$\in$ = Emissivity
σ = Stefan Boltzmann constant

2 Model parameter Identification

In this section, formulation strategy for the model parameter identification is discussed which forms the basis of identifying unknown parameters of TCL system. Consider a mechanistic model given by the following differential equation:

State dynamics:

$$\frac{dX}{dt} = \mathbf{f}(X, U, \theta) \tag{5}$$

Measurement model:

$$Y(t) = CX(t) + v(t) \tag{6}$$

where $\mathbf{X} \in R^n$ represents state variables, $\mathbf{U} \in R^m$ represents manipulated inputs, $\mathbf{Y} \in R^r$ represents measured outputs, $\theta \in R^p$ represents the unknown parameter set, and $v(t)$ represents errors in the measurement and unmeasured output disturbances. The ODE-IVP given by Equation 5 can be integrated to arrive at a non linear discrete time model over the inteval t_k to t_{k+1} with initial condition $X(k)$ under the following given assumption that the manipulated inputs are piecewise constant during the sampling interval. Any numerical computation method can be used to solve Equation 5 and approximate $F(X(k), U(k), \theta)$ giving the following discrete time control model

$$X(k+1) = F(X(k), U(k), \theta) \equiv X(k) + \int_0^T \mathbf{f}(X(\tau), U(k), \theta) d\tau \tag{7}$$

$$Y(k) = CX(k) + v(k) \tag{8}$$

The model parameters of the system are identified by minimizing the innovation given by the objective function (J) defined below in Equation 9

$$J = \underset{\theta}{\arg\min} \sum_{k=1}^{N} v(k)^T R^{-1} v(k) \tag{9}$$

Subject to

$$X(k+1) = F(X(k), U(k), \theta)$$

$$v(k) = Y(k) - CX(k)$$

where R represents covariance of the measurement error $v(k)$. The fixed parameters of the TCL model are given in the Table 1.

Table 1: TCL: Fixed parameters

Quantity	unit	Value
Q_1	W	0 to 1
Q_2	W	0 to 0.75
c_p	J/KgK	500
A	m^2	$1 * 10^{-3}$
A_s	m^2	$2 * 10^{-4}$
m	Kg	0.004
$\in$	$-$	0.9
σ	W/m^2K^4	$5.67 * 10^{-8}$

Table 2: TCL: Tuned parameters

Tuned Parameters	unit	Value
U	W/m^2K	6.7853
α_1	$W/(\%heater)$	0.0050
α_2	$W/(\%heater)$	0.0036

Now, the other environment sensitive model parameters U, α_1, and α_2 are estimated by minimizing the objective function given by Equation 9 discussed in the earlier section. For this purpose, experiments with a sampling time of 4 seconds are performed on TCL. For collecting data in a wide range of operating conditions, PRBS signal is given to the individual heaters with a varying mean of the operating input and corresponding temperature readings are noted in an open loop experiment. This input-output data is used to tune the parameters. Thus the unknown parameter set θ for TCL includes U, α_1 and α_2. For obtaining measurement noise (R) and process noise (Q) covariances, steady state experimental data in the time range of 800 seconds to 1000 seconds as shown in the Figure 2 is considered.

$$R = \begin{bmatrix} 0.0631 & 0 \\ 0 & 0.1346 \end{bmatrix} \tag{10}$$

The objective function is minimized using fmincon solver in MATLAB to obtain the tuned parameters given in Table 2.

The comparison of open loop model simulation and experimental results is shown in the Figure 2 and Figure 3.

It can be seen from Figure 2 that the temperature readings are tracked by the model efficiently and can thus be used for state estimation and control based studies.

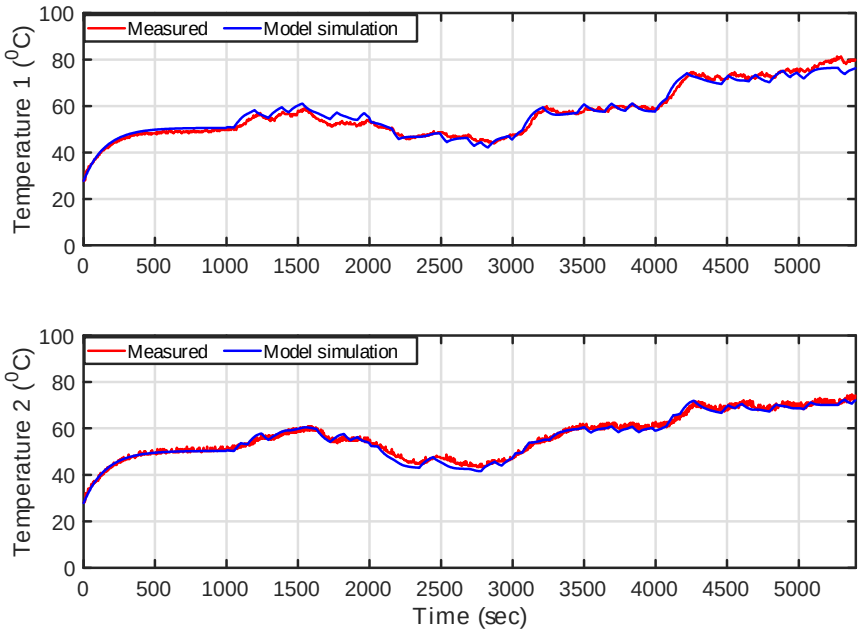

Figure 2: TCL: Open loop model simulation vs measured temperature

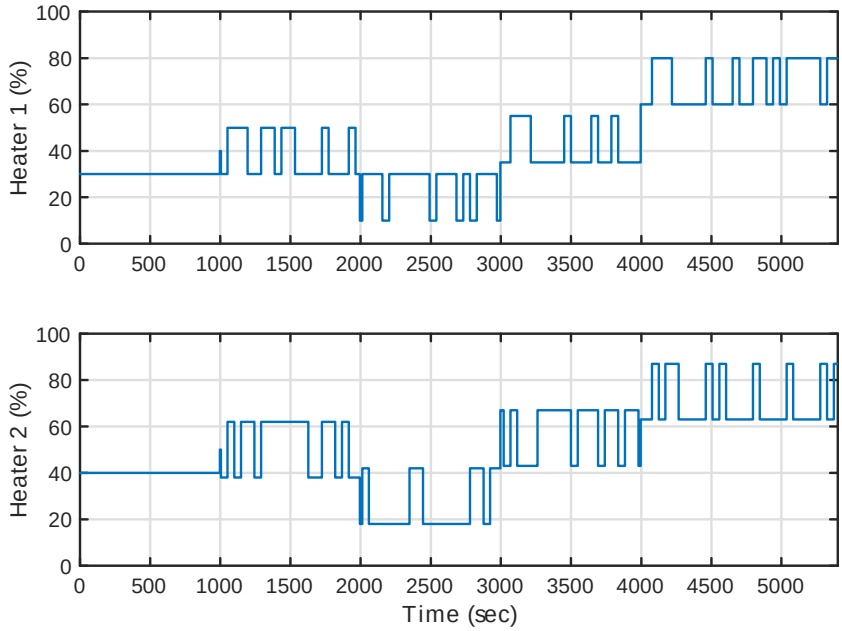

Figure 3: TCL: Open loop heater output